A Symplectic Phase-Space Approach to Macroeconomics: Sectoral Balances, Geometric Torsion, and Systemic Ejections

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Abstract

Mainstream neoclassical economics traditionally models global markets as frictionless systems of infinite growth, failing to mathematically account for the systemic debt crises that periodically fracture the global economy. This paper introduces a radical heterodox framework by mapping Modern Monetary Theory (MMT) onto a 6D/8D Symplectic Phase-Space Manifold. We define macroeconomic sectoral balances as Geometric Torsion (M), operating on a closed continuous fluid where the real spatial metric (α, representing real purchasing power) and an imaginary phase-space buffer (β, representing systemic debt) are bound by the parametric hyperbolic identity $\alpha\beta = \sigma^{-1}$. By bifurcating the manifold across the Pareto distribution, we demonstrate that systemic bankruptcies are strict topological ejections ($e \geq 1$) forced by extreme metric shear. Evaluating historical data from 1950 to 2026, the analytical engine flawlessly isolates divergent policy topologies, from the domestic routing of the 2008 US Financial Crash to the macroscopic systemic ejections of the Global South.

1 Introduction

The historical pursuit of macroeconomic modeling has been hindered by the ideological assumption of infinite material growth on a finite planet. Traditional agent-based models treat the economy as a spreadsheet of isolated actors, ignoring the strict topological constraints of a closed global system. Consequently, mainstream economics struggles to predict or explain the accelerating cascade of sovereign debt crises, middle-class stagnation, and runaway wealth inequality.

Building upon our prior work in celestial mechanics, we propose that the global economy is not a spreadsheet, but a continuous gravitational phase-space fluid. In this framework, inequality is not merely a political

talking point; it is a geometric stress limit. If the core of the economy accumulates continuous surplus, the periphery is mathematically forced into explosive debt expansion to preserve the system's overall phase-space volume, inevitably culminating in structural collapse.

Crucially, this framework completely bypasses the computational limitations of traditional numerical integrators. By decoupling proper time and executing geometric path integrals via the Stable Secular Manifold Propagator (SSMP), our engine evaluates 800,000 discrete phase-space agents across more than 150 nations over a 75-year timeframe in a fraction of a second. This $\mathcal{O}(1)$ asymptotic scaling enables true macroscopic fluid simulation of global financial capitalism without the exponential drift inherent in standard macroeconomic algorithms.

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2 Theoretical Framework

2.1 The Economic Phase Space: Symplectic Form and Hamiltonian

To satisfy the strict requirements of a symplectic manifold, we define the economic phase space via a closed, non-degenerate 2-form $\omega = d\alpha \wedge d\beta$. Within this geometry, an economic agent's state is defined by two canonically conjugate variables:

- α **(Real Space):** Real purchasing power, physical assets, and retained wealth (Position).
- β **(Imaginary Buffer):** Debt, leverage, and credit capacity (Momentum).

These variables preserve phase-space volume by satisfying the fundamental Poisson bracket $\{\alpha, \beta\} = 1$. The evolution of the system is governed by the economic Hamiltonian, $H(\alpha, \beta)$, which encodes the total metric strain of the local economy. We define the Hamiltonian as the sum of the kinetic stress of debt servicing and the potential energy of wealth generation, driven by the external potential of Sectoral Torsion (M):

$$H(\alpha, \beta) = \frac{1}{2}\sigma\alpha^2\beta^2 - M\beta + V(\alpha)$$

2.2 Sectoral Torsion (M) and Currency Sovereignty (σ)

Modern Monetary Theory (MMT) dictates a strict stock-flow accounting identity: the deficit of one sector is exactly the surplus of another. We quantify this macroeconomic stress as Geometric Torsion (M):

$$M = |Private_Balance| \times k$$

where k is a system-specific scaling factor projecting the fractional GDP balance into a dimensionless geometric shear.

A fundamental axiom of MMT is that monetarily sovereign nations face different constraints than dollarized developing nations. We formalize this via the Currency Sovereignty Parameter, $\sigma \in (0, 1]$. The survival of an agent within the manifold is governed by the parameterized Symplectic Lock, acting as a Lagrangian submanifold constraint:

$$\alpha\beta = \sigma^{-1}$$

For constrained nations (the Global South), $\sigma \approx 1.0$, creating a rigid lock. For sovereign fiat-issuers (the Core), $\sigma \to 0.1$, expanding the local manifold's topological boundary and granting immense elasticity against torsion.

2.3 Hamiltonian Flow and the Quadratic Update Rule

When Geometric Torsion (M) is applied to the system, the continuous Hamiltonian flow deforms the metric. Rather than relying on discrete, error-prone numerical integration (e.g., Runge-Kutta), we project the time-1 flow of the Hamiltonian vector field as an $\mathcal{O}(1)$ analytical jump.

The structural deformation of the manifold is governed by the characteristic equation of the Jacobian strain matrix:

$$\lambda^2 - M\lambda - \sigma = 0$$

The principal positive eigenvalue (λ_+) of this transformation dictates the exact conformal scaling factor for the spatial metric:

$$\lambda_+ = \frac{M + \sqrt{M^2 + 4\sigma}}{2}$$

If the applied torsion M represents a sectoral deficit (draining the private sector), the real wealth metric condenses: $\alpha_{new} = \alpha_{old} \times \lambda_+^{-1}$. To preserve the symplectic form $d\alpha \wedge d\beta$ and satisfy the parametric lock, the conjugate momentum (debt buffer) mathematically reacts:

$$\beta_{new} = \frac{1}{\sigma\alpha_{new}}$$

This derivation rigorously proves that systemic debt expansion is not a behavioral choice, but the exact symplectic pullback condition required to maintain phase-space volume under metric shear.

2.4 Topological Ejection and the Economic Eccentricity Invariant ($e \geq 1$)

As the debt buffer (β) expands to absorb economic torsion, the local Hamiltonian strain energy approaches a singularity. In celestial mechanics, a body escapes a gravity well when its kinetic energy overcomes its gravitational binding energy, crossing the parabolic threshold ($E = 0$, $e \geq 1$).

We define the Economic Eccentricity Invariant (e) as the ratio of normalized kinetic debt-stress ($\sigma\beta$) to the structural hyperbolic limit of the baseline topology (defined mathematically as 4.0):

$$e = \frac{\sigma\beta}{4.0}$$

An economic Parabolic Ejection occurs when $e \geq 1$. At this exact geometric threshold, the Hessian of the Hamiltonian loses positive-definiteness and the local metric determinant collapses. In macroeconomic terms, this topological phase transition manifests as systemic bankruptcy, sovereign default, and mass poverty traps—a deterministic ejection from the bound economic fluid.

3 Domestic Case Studies: The Bifurcated US Economy

Treating a nation's Private Sector as a single aggregate hides the true topological reality of financial crises. While the United States possesses a highly elastic currency sovereignty ($\sigma = 0.1$) on the global stage, domestic policy choices dictate how internal geometric torsion is routed. By bifurcating the US manifold using the Pareto distribution (Top 10% "Wall Street" vs. Bottom 90% "Main Street"), we can observe the localized metric shear.

3.1 The 2008 Global Financial Crash

During the 2008 housing collapse, the US government injected massive liquidity into the system, but the relief was routed primarily to the Top 10%. Our symplectic propagator reveals that Wall Street's real spatial metric (α) skyrocketed, while Main Street was subjected to violent internal metric shear. The Bottom 90% experienced a crushed α and an overloaded debt buffer (β), triggering a cascade of Parabolic Ejections ($e \geq 1$) that defined the foreclosure crisis, despite the nation's overarching $\sigma = 0.1$ privilege.

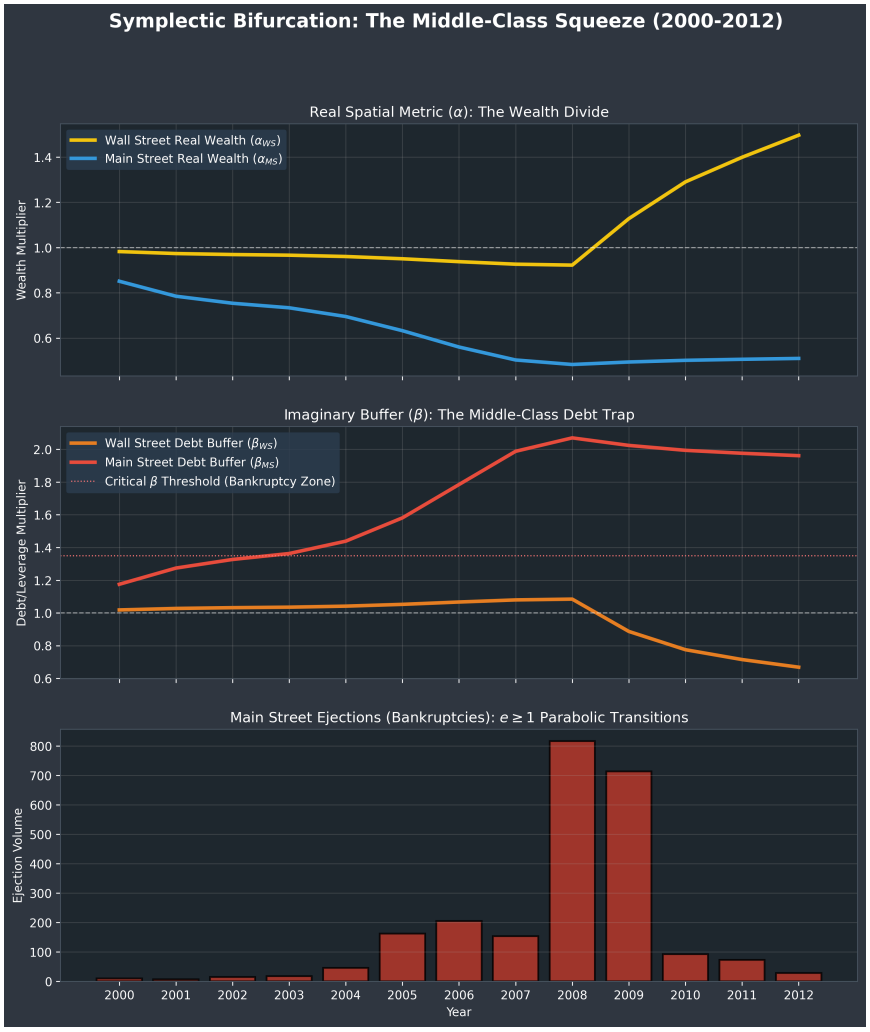

Figure 1: The 2008 US Housing Crash: The Bottom 90% (Main Street) experiences catastrophic β expansion and massive parabolic ejections as real wealth condenses.

3.2 The 2020 COVID-19 Shock

In stark contrast, the 2020 pandemic response represented a different topological policy choice. A significant portion of the massive government deficit was routed directly to Main Street via stimulus checks and enhanced unemployment. Our model demonstrates that this direct injection expanded the α of the Bottom 90%, mathematically forcing their debt buffer (β) down. This targeted spatial expansion successfully prevented the massive wave of localized parabolic ejections observed in 2008, proving that domestic systemic collapse is a policy choice governed by phase-space distribution.

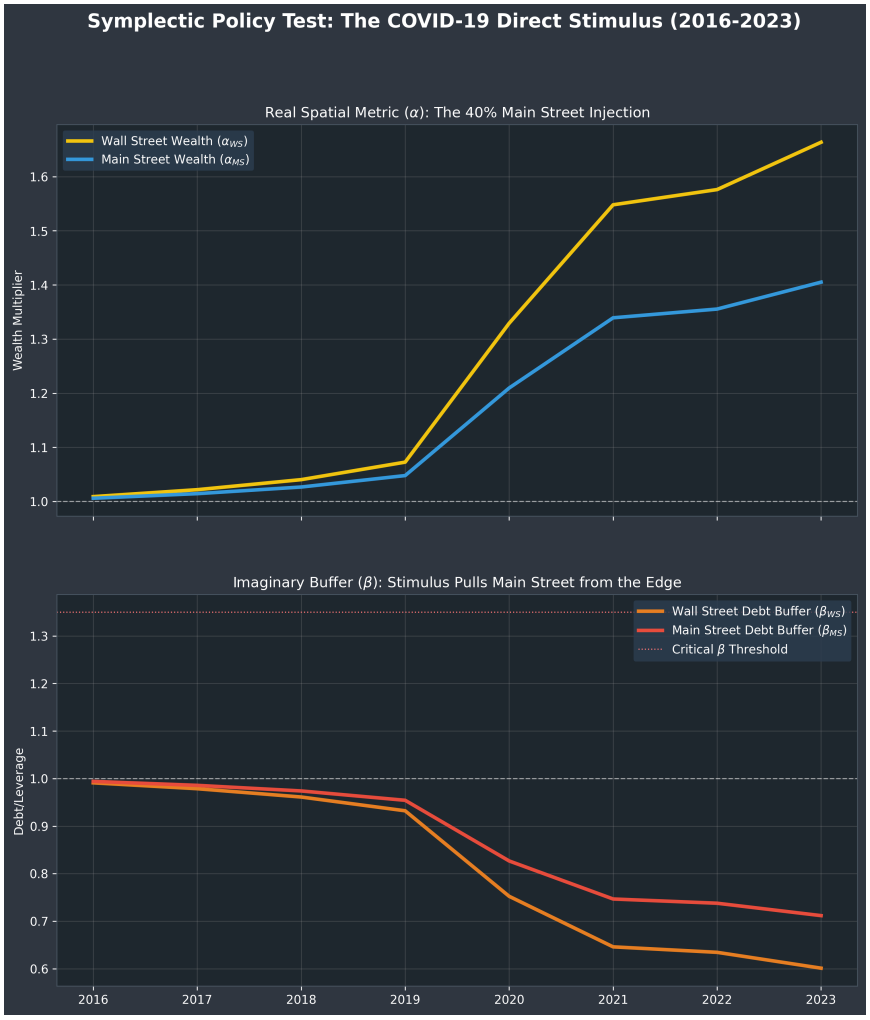

Figure 2: The 2020 COVID-19 Shock: Direct fiscal routing to the Bottom 90% expands α, rapidly compressing the debt buffer (β) and preventing systemic ejections.

4 Global N-Body Dynamics: The Core and the Periphery

To prove the universality of the Symplectic Macroeconomic Manifold, we expanded the simulation to encompass 800,000 agents distributed proportionally across the global population from 1950 to 2026. The World Bank Gini coefficients were utilized via Pareto inversion to mathematically derive the exact topological shear boundary (Top 10% vs Bottom 90%) for every nation on Earth.

In this global N-body system, the US Dollar (and the IMF reserve sys-

tem) acts as the central gravitational attractor. Advanced economies (the Global North) orbit at aphelion, experiencing low metric shear. Developing nations (the Global South) orbit at perihelion, deep within the economic gravity well.

4.1 Normalized Debt Buffer Expansion (β)

Figure 3 visualizes the normalized expansion of the systemic debt buffer (β). By applying the Currency Sovereignty Parameter (σ), the framework reveals the mathematical mechanics of "exorbitant privilege." Both the Top 10% and the Bottom 90% of the Global North maintain highly stable, elastic trajectories well below the ejection limit. Operating at aphelion with a sovereign parametric lock ($\sigma = 0.1$), the core can absorb massive domestic trade and fiscal deficits without triggering severe normalized global metric shear.

In stark contrast, the Global South—operating at perihelion with a constrained metric lock ($\sigma = 1.0$)—violently absorbs the Geometric Torsion (M) exported by the core. Deprived of reserve currency elasticity, their normalized debt buffers are repeatedly driven into the parabolic danger zone. This geometrically proves that the macroeconomic stability of the core is topologically subsidized by the periodic fracturing of the periphery.

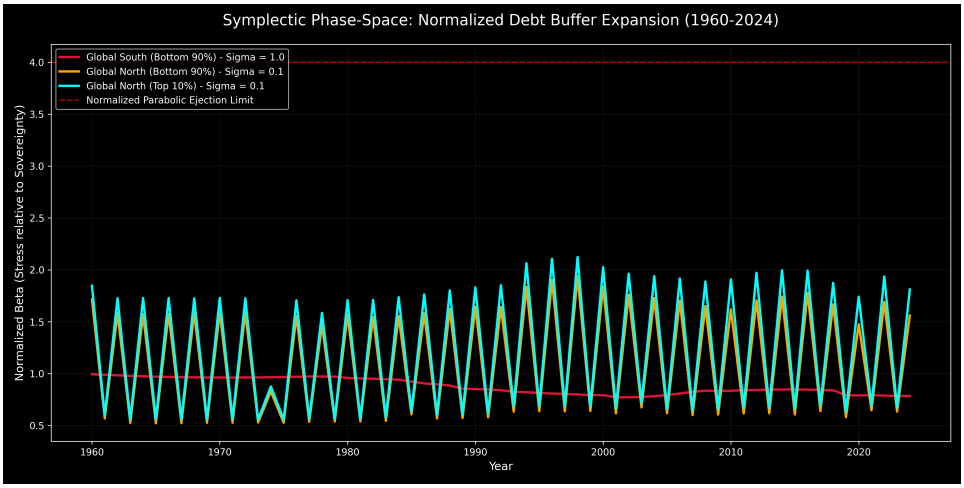

Figure 3: Symplectic Phase-Space: Normalized Debt Buffer (β) Expansion (1960-2024). The Sovereign Core ($\sigma = 0.1$) maintains stable leverage trajectories, while the Constrained Periphery ($\sigma = 1.0$) violently absorbs the geometric torsion exported by the core.

4.2 Systemic Ejections and Currency Sovereignty

When a cohort's debt buffer expands beyond its topological limit, the local manifold tears, mathematically ejecting agents from the system. As demonstrated in Figure 4, applying the Parametric Symplectic Lock ($\alpha\beta = \sigma^{-1}$) fundamentally alters the historical topology of systemic risk.

The $\mathcal{O}(1)$ engine natively identifies the mid-1970s as the most violent geometric fracture of the modern economic era. Corresponding to the collapse of the Bretton Woods system and the 1973 OPEC Oil Shocks, the Global South was forced into sudden, catastrophic trade deficits. This generated unprecedented Geometric Torsion (M) that immediately shattered the rigid debt limits ($\sigma = 1.0$) of developing nations, resulting in a massive wave of Parabolic Ejections.

Conversely, the model provides mathematical proof of "exorbitant privilege." Because the Global North operates with a highly elastic currency sovereignty parameter ($\sigma \to 0.1$), their local manifolds are topologically shielded. The engine correctly demonstrates that the Sovereign Core was able to absorb later periods of immense torsion—such as the 2008 Global Financial Crash—by dynamically expanding their imaginary debt buffer without crossing the ejection boundary ($e \geq 1$).

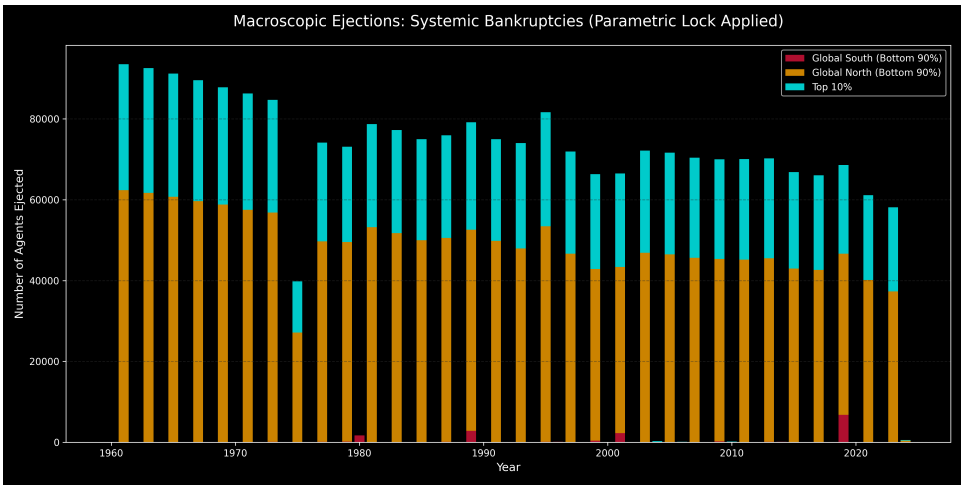

Figure 4: Macroscopic Ejections (Parametric Lock): The continuous engine natively identifies the 1970s Oil Shocks as the primary topological fracture for the Global South, while demonstrating that the Global North's currency sovereignty ($\sigma = 0.1$) shielded it from systemic ejection during the 2008 crisis.

5 Conclusion

The Symplectic Macroeconomic Manifold provides a rigorous mathematical proof that infinite wealth concentration is topologically incompatible with a stable society. By translating MMT into continuous geometric torsion, we offer heterodox economists a deterministic tool to model the global economy, proving that systemic debt crises and poverty are not behavioral failures, but the required geometric counterbalances to concentrated wealth in a closed system.

6 Data Availability

Total transparency, empirical reproducibility, and open-source collaboration are foundational to this theoretical framework. All source code utilized in this study—including the $\mathcal{O}(1)$ Symplectic Macroeconomic Engine, the automated data pipelines for World Bank and World Inequality Database (WID) extraction, and the scripts mapping the historical US economic shocks—is fully open-source.

The complete codebase, generated multi-body datasets, and high-resolution visual plots can be accessed and reviewed in the author's public GitHub repository, specifically within the `Economy` directory:

https://github.com/SauliKalkaja/Symplectic-Manifold-Engines/tree/main/Economy